

1. Our Earth and our Solar System

Heavenly Bodies

- The sun, the moon, the stars, the planets, etc. are all known as heavenly bodies.

Stars

- The heavenly bodies that twinkle are called stars.
- Stars have their own light.
- The sun is also a star. It is closer to us than any of the other stars. Hence, it appears big and brilliant.
- In its bright light, during the day, we cannot see other stars.
- Some stars are prominent (bright) while some are faint (dim).
- Some stars twinkle while some do not.
- Stars are very far away from the earth. In the dark night sky, we can see them clearly.

Planets

- The heavenly bodies that do not twinkle are called planets.
- Planets do not have light of their own. They get light from the sun.
- Planets are comparatively close to the earth, so we can see their round shapes clearly.
- Planets revolve around a star, even as they rotate around themselves.
- The size of planets varies — Jupiter is the largest and Mercury is the smallest.

The Solar System

- Our earth is a planet. It gets its light from the sun.
- Earth moves around the sun. This movement is called the revolution of the earth.

- Besides earth, there are 7 other planets : Mercury, Venus, Mars, Jupiter, Saturn, Uranus and Neptune.
- Every planet revolves around the sun along a specific path called its orbit.
- The sun (a star) and all the planets revolving around it together form the solar system.
- Each planet takes a different time to complete one revolution around the sun.

Satellites

- Some heavenly bodies revolve around planets. These are called satellites.
- Satellites too get their light from the sun.
- The moon is a satellite of the earth. It revolves around the earth.
- Most of the planets in the solar system have satellites.

Dwarf Planets

- There are some smaller heavenly bodies that revolve around the sun. They are called dwarf planets.
- The most well-known dwarf planet is Pluto.
- Dwarf planets revolve independently around the sun. They have an orbit of their own.
- Dwarf planets are smaller in size than regular planets.

Asteroids

- Between Mars and Jupiter, there is a band of numerous small heavenly bodies called asteroids.
- Asteroids also revolve around the sun.
- This region is called the asteroid belt. It covers a large area between the two planets.

Gravity

- All heavenly bodies exert a force of attraction on one another. This is called the force of gravity.

- The sun pulls planets towards it. Planets also tend to move away. These two forces keep a planet in its orbit.
- Similarly, gravity keeps satellites revolving around their planets.
- Due to earth's gravity, everything on earth stays on it. Things thrown upward always fall back down.
- The gravitational pull of the sun acts on all planets in the solar system.

India's Space Missions

1. Chandrayaan-1

- Launched on 22 October 2008 by ISRO to the moon.
- It was an unmanned mission — no people were on board.
- Objective : deeper study of the moon.

2. Mangalyaan (M.O.M / Mars Orbit Mission)

- Launched on 5 November 2013.
- Entered orbit around Mars on 24 September 2014.
- ISRO achieved this in its first attempt — a great achievement!
- Both missions were unmanned — no people on board.
- Objective : deeper study of the moon and Mars.

Quick Revision Points

- * Stars twinkle. Planets do not twinkle.
- * Stars have their own light. Planets get light from the sun.
- * Planets revolve around the sun in a fixed orbit.
- * The moon is a natural satellite of the earth.
- * Dwarf planets (e.g. Pluto) revolve independently around the sun.
- * Asteroids are small bodies found between Mars and Jupiter.
- * Gravity is the force that keeps planets in their orbits.
- * Chandrayaan-1 launched in 2008. Mangalyaan launched in 2013.
- * Both missions were unmanned. Goal : study moon and Mars.